

Disaster Recovery Management Platform

From Excel Chaos to Intelligent DR Coordination

Equity Bank Group — All Subsidiaries

KE · UG · RW · TZ · DRC · SS

Group IT Service Continuity Management (ITSCM)

Version 1.0 | March 2026 | CONFIDENTIAL

Agenda

What We Will Cover Today

01 The Problem

Manual Excel coordination across 100+ critical apps

02 Real Evidence

Finacle TZ DR Exercise — 15 March 2026

03 The Solution

9-module intelligent DR Management Platform

04 System Architecture

Spring Boot · React · PostgreSQL · Redis

05 Workflow Design

Campaign → Runbook → War Room → Report

06 Runbook Enforcement

6-layer enforcement: T-21 through T-1

07 Live War Room

Real-time execution with WebSocket + RTO timer

08 Blocker Intelligence

Carry-forward engine and recurring pattern detection

09 Security Architecture

Banking-grade: LDAP SSO, MFA, AES-256, audit trail

10 Delivery Roadmap

4 phases over 20 weeks — incremental delivery

11 Before vs After

Coordinator impact — day one to go-live

12 Next Steps

Decision points and immediate actions required

The Problem We Are Solving

DR Coordination Today — Manual, Fragile, Invisible

Current State:

- **One Coordinator** — Managing 100+ mission-critical apps alone, every quarter
- **Excel Spreadsheets** — Multiple disconnected trackers — no single source of truth
- **WhatsApp & Email** — DR coordination via informal channels — nothing auditable
- **Zero Enforcement** — App owners ignore runbook requests — no system consequences
- **Lost History** — Blockers from Q1 DR are forgotten by Q3 — same issues repeat
- **Manual Reports** — Post-DR Word docs assembled hours after the exercise ends
- **No Visibility** — Management has no real-time view of DR readiness or execution

100+

Mission-Critical

Applications

6

Group Subsidiaries

KE·UG·RW·TZ·DRC·SS

1

Coordinator

Managing Everything

0

Automated

Enforcement Today

Real Evidence — This Is Not Hypothetical

Finacle Core & Treasury DR Exercise — Tanzania | 15 March 2026 | 22:00 hrs

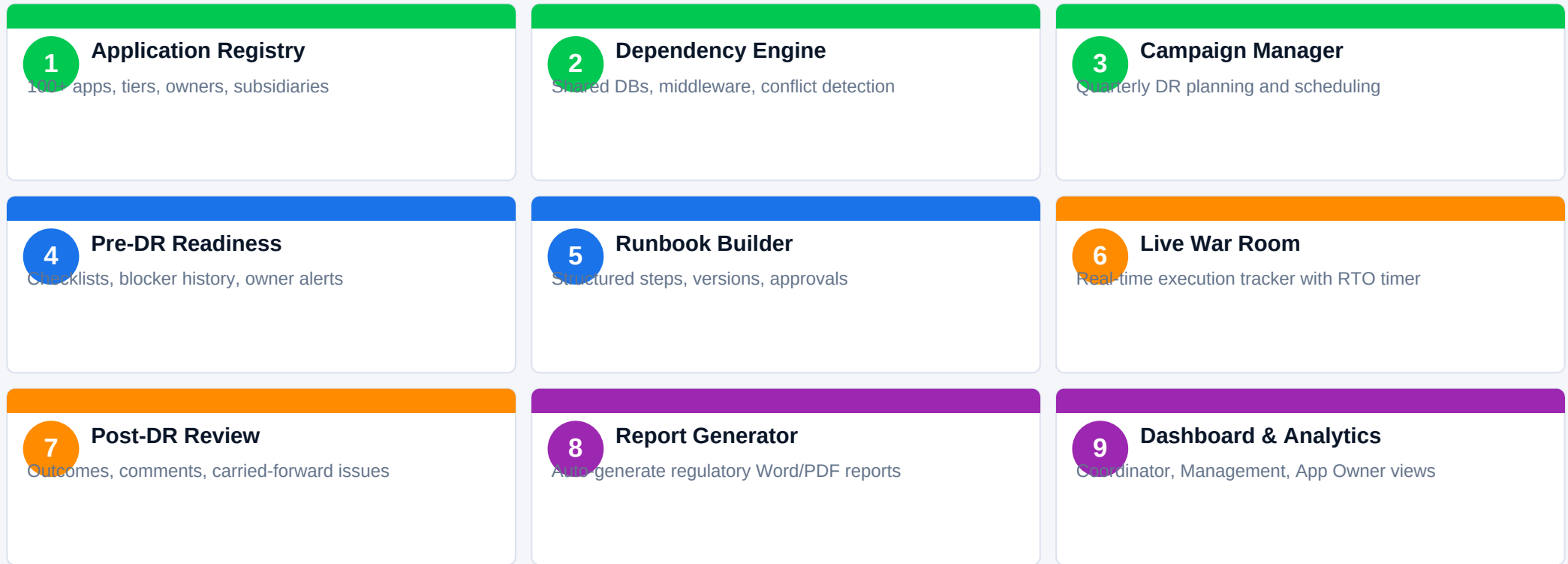
Six distinct failure categories recorded during a real DR exercise — all preventable with the platform:

#	Issue Observed	Root Cause	Platform Solution
1	DBA, DevOps, NOC absent at start	No mandatory attendance enforcement	War Room: mandatory check-in gate
2	Treasury IP config wrong after failover	Manual config — no automation	Runbook step with auto-validation
3	WAY4 RC 907/904 errors post-failover	No structured post-failover checklist	Validation phase in runbook engine
4	Oracle DG TNS entries not updated	Manual file editing across servers	Shared step owned by DBA — tracked
5	NOC TZ DR monitoring not visible	DR env not in monitoring dashboard	Post-DR outcome: NOC action item
6	External dependency (DNB) failure untracked	No third-party dependency mapping	Dependency Engine: external mapping

Every one of these issues would have been caught, assigned, and tracked automatically. None required new hardware — only a better process.

The Solution — 9 Core Modules

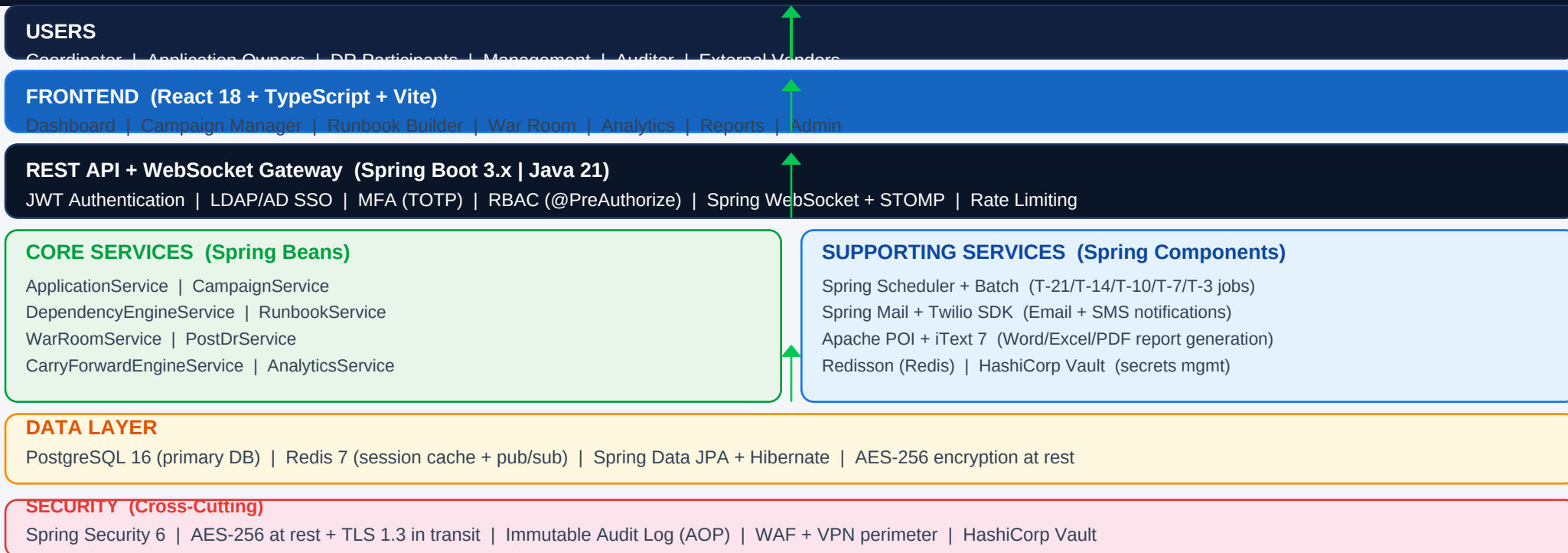
One Intelligent Platform Replacing All Manual Coordination



All 9 modules share one database, one login, one audit trail — working together from day one.

System Architecture

Java 21 + Spring Boot 3.x | React 18 | PostgreSQL 16 | Redis 7



Technology Stack

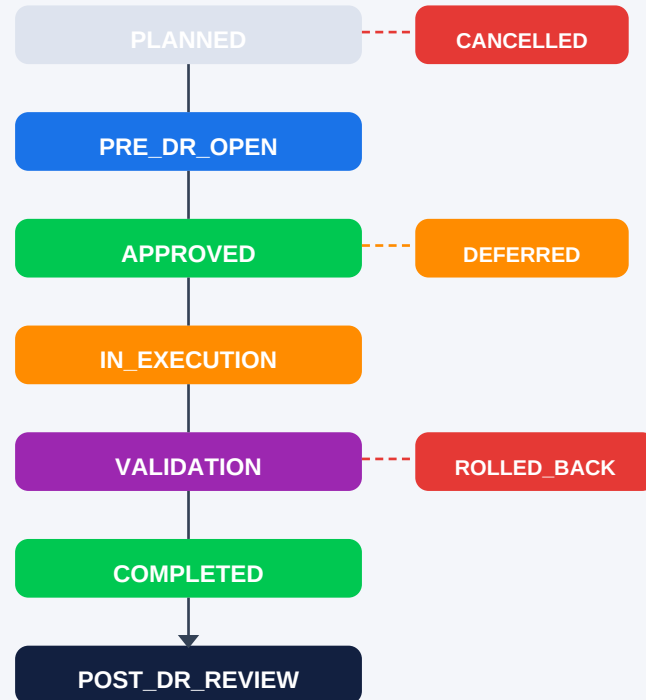
Enterprise Java — Battle-Tested in Banking Environments

Layer	Technology	Rationale
Frontend	React 18 + TypeScript + Vite	Modern, type-safe UI — fast hot-reload dev experience
Backend API	Java 21 LTS + Spring Boot 3.x	Enterprise-grade, banking industry standard, native NetBeans support
Security	Spring Security 6	LDAP/AD SSO, JWT (15-min), MFA (TOTP), RBAC, method-level security
ORM	Spring Data JPA + Hibernate	Type-safe queries, transactional integrity, audit entity support
Real-Time	Spring WebSocket + STOMP	War Room live execution — no external library needed
Background Jobs	Spring Scheduler + Spring Batch	T-21/T-14/T-10/T-7/T-3 reminders, carry-forward engine
Report Engine	Apache POI (Word/Excel) + iText 7 (PDF)	Java-native document generation, digital signing, watermarking
Notifications	Spring Mail + Twilio SDK	Email via corporate SMTP; SMS via Twilio for critical alerts
Database	PostgreSQL 16	JSONB, row-level security, partitioned audit table
Cache/Pub-Sub	Redis 7 + Redisson	Session cache, War Room broadcasting, rate limiting
Secrets	HashiCorp Vault + Spring Cloud Vault	Zero secrets in code or config files — ever
Build / Deploy	Maven + Docker + Docker Compose	Consistent environments — dev, staging, production

Workflow Design

Every DR Activity Follows a Formal State Machine — No Stage Can Be Skipped

Campaign Application Lifecycle



Automated Workflow Triggers

Campaign Created	Checklist items auto-generated for all app owners
T-21 Days	Friendly email reminder to all application owners
T-14 Days	Urgent email if runbook not yet submitted
T-10 Days	Email + SMS to owner + coordinator flagged
T-7 Days (HARD GATE)	Campaign blocked; manager escalation; AT RISK badge
T-3 Days	Status -> CANNOT PROCEED; exception must be documented
T-1 Day (WAR ROOM GATE)	War Room launch disabled without approved runbook
Blocker Raised	Immediate alert to coordinator + line manager
War Room Step Blocked	STOMP broadcast + escalation contact notified
Campaign Closed + Issues	Issues auto-tagged CARRIED_FORWARD to next campaign

Runbook Enforcement — The 6-Layer System

Preparation Is Easy, Visible, and Unavoidable — System Enforces, Not the Coordinator



Six Enforcement Layers

1

Auto-Timeline
Deadlines auto-calculated from DR date — visible the moment the campaign is created

2

Progressive Notifications
Pressure escalates: T-21 friendly → T-14 urgent → T-10 warning+SMS → T-7 manager

3

Hard Workflow Gates
Campaign cannot reach APPROVED; War Room cannot launch without approved runbook

4

Smart UX
Pre-populated from last runbook (35-min review vs 3-hr write) + quality validator

5

Coordinator Dashboard
Live runbook status grid for all apps — one-click Remind or Escalate

6

Social Accountability
Manager escalation at T-7; peer reviewer on hook; management shows readiness %

Live War Room — Real-Time DR Execution

Spring WebSocket + STOMP | RTO Countdown | Shared Steps | Issue Broadcasting

How the War Room Works

- 1

Pre-Launch Check
Coordinator verifies all runbooks are APPROVED & LOCKED. System validates dependencies. Re-authentication required to launch.
- 2

Steps Auto-Loaded
All runbook phases and steps load from the immutable approved snapshot. Owners pre-assigned to each step.
- 3

RTO Timer Starts
Redis-backed countdown begins the moment the session goes ACTIVE. Displayed live to all participants.
- 4

Step Execution
Each step owner marks IN_PROGRESS then DONE or BLOCKED. Real-time updates broadcast via STOMP to all participants.
- 5

Shared Steps
Steps shared across apps (e.g. Oracle DG switchover) executed once by DBA — auto-marks complete on all linked runbooks.
- 6

Issue Logging
HIGH/CRITICAL issues instantly broadcast to all. Escalation contact paged automatically. Every action timestamped.
- 7

Validation Phase
Post-execution — app owners submit business validation results. System holds status until all validations pass.
- 8

War Room Closed
Session sealed. All data locked for audit. Post-DR outcomes board opens automatically for each application.

WAR ROOM — Q1 2026 | Finacle Core TZ

RTO TIMER

01:42:15

CONNECTED: App Owner (2) DBA (1) NOC (1) Infra (1) Coordinator (1)

Phase	Step	Owner	Status
Pre-Cut	Stop App Server	App Owner	DONE
Pre-Cut	Backup config files	App Owner	DONE
Cutover	Oracle DG Switchover	DBA	IN PROGRESS
Cutover	Update TNS Entries	DBA	PENDING
Cutover	Restart App (DR)	App Owner	PENDING
Validate	Business Transaction Test	App Owner	PENDING

Without this engine: Q1 2026 blockers are forgotten. Q3 2026 — same blockers appear. Same people apologise. Same issues persist.

What the Carry-Forward Engine Does

Spring Batch Job

Runs nightly after campaign close — scans all unresolved checklist items and war room issues

Automatic Linking

Every carried-forward item links to its parent — full history visible: 'this issue first appeared Q3 2024'

Age Tracking

Each issue carries a 'cycles_unresolved' counter — items unresolved for 3+ cycles flagged CHRONIC

Owner Notification

App owner receives a carried-forward summary email at start of next campaign — no surprises

Coordinator View

Dashboard shows: 20 carried-forward items this cycle, 5 chronic (3+ cycles), 2 new SYSTEMIC_RISK

Audit Trail

Every carry-forward link recorded with actor, timestamp, and resolution status in immutable audit log

Pattern Detection — SYSTEMIC_RISK Alerts

SYSTEMIC_RISK Flag

If the same issue category appears in 3 or more consecutive DR cycles, the engine raises a SYSTEMIC_RISK alert — visible to coordinator AND management dashboard

Category Analysis

Issues are tagged by category: INFRASTRUCTURE, APPLICATION, DATABASE, PROCESS, EXTERNAL. The engine detects patterns per category per application

Management Escalation

SYSTEMIC_RISK items automatically appear on the management dashboard with full history — prompts permanent fix, not just workaround

Resolution Tracking

When a SYSTEMIC_RISK item is finally resolved, the platform records the resolution and removes the flag — with full audit trail

Banking-Grade Security Architecture

CBK IT Risk | ISO 27001 | PCI-DSS | BCBS 239 | Zero-Trust Design

PERI PERIMETER WAF + VPN ● Web Application Firewall (OWASP CRS) ● VPN-only access from corporate network ● DDoS protection on ingress ● Bot detection and IP allowlisting CBK IT Risk Guidelines	IDEN IDENTITY Auth & Access ● LDAP/AD SSO — one login, no new password ● MFA (TOTP) mandatory for privileged roles ● JWT: 15-min access, 8-hr refresh tokens ● RBAC: 8 roles, method-level @PreAuthorize Bank of Tanzania IT Regulations	APPL APPLICATION API Security ● OWASP Top 10 mitigations built-in ● CSRF tokens on all state-changing endpoints ● Input validation + parameterised JPA queries ● Rate limiting per user/IP via Redis ISO/IEC 27001:2022	DATA DATA Encryption ● AES-256 encryption at rest (sensitive fields) ● TLS 1.3 in transit — no HTTP downgrade ● HashiCorp Vault — zero secrets in code ● PostgreSQL row-level security PCI-DSS v4.0	AUDI AUDIT Immutable Log ● Every action logged via AOP (AuditLogAspect) ● Audit table: INSERT ONLY — no UPDATE/DELETE ● Actor, IP, timestamp, before/after values ● Export to SIEM — CBK/ISO 27001 aligned BCBS 239
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DR Activity Types Supported

10 Distinct Scenarios — Each with Purpose-Built Workflow and Runbook Template

FULL_FAILOVER Full Failover DC App + DC DB migrate to DR App + DR DB. Complete site failover. Highest complexity.	APP_ONLY App-Only Failover DC App moves to DR App. Database stays in primary DC. Typical for web-tier failover.
DB_ONLY DB-Only Failover Application stays in DC. Database switches to DR via Oracle Data Guard activation.	APP_REPOINT App Repoint DC App is reconfigured to point to DR database. No application movement required.
PARTIAL Partial Failover A defined subset of services fails over while others remain in DC. Complex routing.	ACTIVE_ACTIVE Active-Active Both DC and DR sites serving live traffic simultaneously. Load-balanced operations.
ROLLBACK_DRILL Rollback Drill Structured test of the return-to-DC (fallback) procedure after a previous failover.	COLD_STANDBY Cold Standby Activation DR environment was powered off. Platform guides warm-up, boot sequence, validation.
TABLETOP Tabletop / Simulated Walkthrough only — no system movement. Role-playing the DR scenario for training.	EXTENDED_OPS Extended DR Operations Running from DR site for 30+ days. Platform tracks extended ops checklist and RTO.

Delivery Roadmap — 4 Phases

Incremental Delivery — Value from Week 6, Full Platform by Week 20+

Phase 1 Foundation Weeks 1–6	Phase 2 Execution Engine Weeks 7–12	Phase 3 Intelligence Weeks 13–20	Phase 4 Advanced Weeks 21+
● Week 1: Auth, LDAP SSO, MFA, Audit Log ● Week 2: Application Registry (import Excel) ● Week 3: Dependency Engine + Conflict Detection ● Week 4: Campaign Manager + DR Scheduling ● Week 5: Pre-DR Checklist + Blocker Tracking ● Week 6: Coordinator Dashboard	● Week 7: Runbook Builder (drag-and-drop steps) ● Week 8: Runbook Approval + T-21 Enforcement ● Week 9: Live War Room (WebSocket + RTO timer) ● Week 10: Issues Module + STOMP Broadcasting ● Week 11: Post-DR Outcome Board ● Week 12: Blocker Intelligence + Carry-Forward	● Week 13–14: Email + SMS Notification Engine ● Week 15: Action Item Tracker ● Week 16: Auto Report Generator (Word + PDF) ● Week 17: Management Dashboard + Charts ● Week 18: App Owner Self-Service Portal ● Week 19–20: Dependency Graph + Analytics	● ServiceNow Change Request integration ● Automated Risk Scoring Engine ● Regulatory Evidence Pack (ZIP + signed PDFs) ● External Vendor Access (time-limited JWT) ● Mobile War Room (responsive Tailwind) ● NOC Integration + monitoring dashboard

Week 6 Excel Tracker Replaced	Week 9 First Live War Room	Week 12 Intelligent Blocker Tracking	Week 16 Auto Report Generation	Week 20 Full Platform Live
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Before vs After — What Changes for the Coordinator

From Manual Chaos to Intelligent Automation

BEFORE (Manual Today)	AFTER (With the Platform)
● Email each app owner individually	● System auto-sends at T-21, T-14, T-10, T-7, T-3 with escalation
● Chase blockers via WhatsApp and phone calls	● Blockers visible on dashboard; auto-escalates to manager at T-7
● Maintain Excel tracker per DR cycle	● All 100+ apps, schedules, and statuses in one real-time screen
● Manually check who has filled runbooks	● Live runbook status grid — one-click Remind or Escalate per app
● Run DR from static Word/Excel runbook	● Live War Room — steps auto-loaded, RTO countdown, real-time tracking
● Take notes on issues during the call	● Issues logged inline, categorised, assigned, and carried forward
● Manually write post-DR Word report (hours)	● Report auto-generated from execution data in under 2 minutes
● Forget Q1 blockers before Q3 DR exercise	● Carried-forward blockers appear prominently — age and cycle tracked
● One person managing everything manually	● App owners are accountable; coordinator oversees, not chases
● No audit trail for compliance purposes	● Every action immutably logged with actor, IP, timestamp, old/new values

VS

Next Steps

Actions Required to Move from Proposal to Delivery

